The relationship between crime and wealth per municipality in the Netherlands

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Set-up your environment

Reproducibility

This project uses `renv` for package management. To restore the environment, run in the Console:renv::restore()

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Part 1 - Identify a Social Problem

"*Dutch municipalities with higher income inequality have a higher number of recorded violent crimes per 1,000 inhabitants*"

1.1 Describe the Social Problem

Crime and socioeconomic inequality are important social issues in the Netherlands. Municipalities differ in terms of household wealth and the number of registered crimes. Previous research suggests that lower levels of wealth may be associated with higher crime rates because economic disadvantages can lead to social exclusion, unemployment, and fewer opportunities. Understanding this relationship is important for policymakers and local governments, as it can help them develop more effective strategies to reduce crime and improve social conditions. This study investigates whether Dutch municipalities with lower household wealth experience higher levels of registered crime per 1,000 households between 2022 and 2024 (Nieuwbeerta et al, 2008).

Part 2 - Data Sourcing

2.1 Load in the data

To reproduce, download the datasets below and save them in the `data/` folder of the project.

Dataset	Source	Link
misdrijven_2022_2023.csv	Data Politie	https://data.politie.nl/#/Politie/nl/dataset/47013NED/table?ts=1780489389758
Misdrijven_2024.csv	Data Politie	https://data.politie.nl/#/Politie/nl/dataset/47013NED/table?ts=1780489389758
vermogen_2022.csv	CBS StatLine	https://opendata.cbs.nl/#/CBS/nl/dataset/86160NED/table?ts=1780405280895
vermogen_2023.csv	CBS StatLine	https://opendata.cbs.nl/#/CBS/nl/dataset/86160NED/table?ts=1780405280895
Vermogen_huishoudens_2024.csv	CBS StatLine	https://opendata.cbs.nl/#/CBS/nl/dataset/86160NED/table?ts=1780405280895

Library

2.2 Dataset description

This project uses two public datasets. The crime data was collected by the Dutch Police (Data Politie) and covers recorded crimes per municipality for 2022, 2023, and 2024. The wealth data was collected by Statistics Netherlands (CBS) and measures household wealth per municipality for the same years. Both datasets are available at the municipal level (*gemeente*), making them suitable for comparison across regions. A key limitation of the crime data is that it only includes **recorded** crimes — unregistered crimes are not captured, which may underrepresent the true crime rate. The wealth data is derived from tax records, which may underrepresent informal income sources.

2.3 Describe the type of variables included

The dataset contains both socioeconomic status (SES) information and crime-related information. The main SES variable is household wealth, measured by average and median household wealth per municipality. The second key variable is the number of registered crimes per municipality. The data does not contain individual health information. All data comes from administrative sources collected by the Dutch Central Bureau of Statistics (CBS) rather than interviews or surveys. The unit of observation is the municipality, which means that each observation represents a Dutch municipality rather than an individual person or household.

Part 3 - Quantifying

3.1 Data cleaning

The raw dataset contained missing values coded as `"."` and numeric columns stored as text with commas as decimal separators. The following steps were applied: - Missing values (`"."`) were replaced with `NA` -

Column names were renamed to readable English variable names - Numeric columns were converted from text to numeric format - Rows with missing wealth values were removed, as these represent non-municipal aggregates (e.g. national totals)

Vermogen 2022

```
vermogen_2022_clean <- vermogen_2022 %>%
  mutate(across(everything(), ~ na_if(., "."))) %>%
  rename(
    gemeenten            = `Regio's`,
    huishoudens_x1000    = `x 1 000`,
    percentage           = `%`,
    vermogen_mld         = `mld euro`,
    vermogen_per_persoon = `1 000 euro...5`,
    vermogen_mediaan     = `1 000 euro...6`
  ) %>%
  mutate(across(c(huishoudens_x1000, vermogen_mld, vermogen_per_persoon, vermogen_mediaan),
              ~ as.numeric(str_replace(., ",", ".")))) %>%
  filter(!is.na(vermogen_mld))
```

Vermogen 2023

```
vermogen_2023_clean <- vermogen_2023 %>%
  mutate(across(everything(), ~ na_if(., "."))) %>%
  rename(
    gemeenten            = `Regio's`,
    huishoudens_x1000    = `x 1 000`,
    percentage           = `%`,
    vermogen_mld         = `mld euro`,
    vermogen_per_persoon = `1 000 euro...5`,
    vermogen_mediaan     = `1 000 euro...6`
  ) %>%
  mutate(across(c(huishoudens_x1000, vermogen_mld, vermogen_per_persoon, vermogen_mediaan),
              ~ as.numeric(str_replace(., ",", ".")))) %>%
  filter(!is.na(vermogen_mld))
```

Vermogen 2024

```
vermogen_2024_clean <- vermogen_2024 %>%
  mutate(across(everything(), ~ na_if(., "."))) %>%
  rename(
    gemeenten            = `Regio's`,
    huishoudens_x1000    = `x 1 000`,
    percentage           = `%`,
    vermogen_mld         = `mld euro`,
    vermogen_per_persoon = `1 000 euro...5`,
    vermogen_mediaan     = `1 000 euro...6`
  ) %>%
```

```
  mutate(across(c(huishoudens_x1000, vermogen_mld, vermogen_per_persoon, vermogen_mediaan),
                ~ as.numeric(str_replace(., ",", ".")))) %>%
  filter(!is.na(vermogen_mld))
```

Misdrijven 2022 2023

```
misdrijven_2022_2023_clean <- misdrijven_2022_2023 %>%
  mutate(across(everything(), ~ na_if(., "."))) %>%
  rename(
    gemeenten = `Regio's`,
    misdrijven_2022 = `aantal...2`,
    misdrijven_2023 = `aantal...3`
  )
misdrijven_2022_2023_clean <- misdrijven_2022_2023_clean %>%
  mutate(across(c(misdrijven_2022, misdrijven_2023), ~ as.numeric(str_replace(., ",", ".")))) %>%
  filter(!is.na(misdrijven_2022))
```

Misdrijven 2024

```
 misdrijven_2024 <- misdrijven_2024 %>%
  mutate(`Geregistreerde misdrijven` = replace(`aantal`, `aantal` == ".", NA))

misdrijven_2024_clean <- misdrijven_2024 %>%

  mutate(`aantal` = na_if(`aantal`, ".")) %>%
  rename(`Misdrijven 2024` = `aantal`) %>%
  rename(`gemeenten` = `Regio's`)
```

3.2 Generate necessary variables

Variable 1

Each municipality is categorized into one of three wealth groups based on average wealth per person: Low (below €300), Medium (between €300 and €500), or High (above €500). This is done for 2022, 2023, and 2024.

```
summary(vermogen_2022_clean$vermogen_per_persoon)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   125.3   283.8   349.8   384.2   444.6  1898.5
```

```
summary(vermogen_2023_clean$vermogen_per_persoon)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   129.8   295.8   365.7   397.6   456.6  2049.6
```

```r
summary(vermogen_2024_clean$vermogen_per_persoon)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   130.1   300.8   376.8   401.9   472.9  1931.5
```

```r
vermogen_2022_var1 <- vermogen_2022_clean %>%
  mutate(vermogen_groep = case_when(
    vermogen_per_persoon < 300  ~ "Low",
    vermogen_per_persoon < 500  ~ "Medium",
    TRUE                        ~ "High"
  ))

vermogen_2023_var1 <- vermogen_2023_clean %>%
  mutate(vermogen_groep = case_when(
    vermogen_per_persoon < 300  ~ "Low",
    vermogen_per_persoon < 500  ~ "Medium",
    TRUE                        ~ "High"
  ))

vermogen_2024_var1 <- vermogen_2024_clean %>%
  mutate(vermogen_groep = case_when(
    vermogen_per_persoon < 300  ~ "Low",
    vermogen_per_persoon < 500  ~ "Medium",
    TRUE                        ~ "High"
  ))
```

Variable 2

To compare municipalities fairly, we calculate crimes per 1000 households instead of raw totals. Otherwise large cities always score higher due to the higher number of people living there.

```r
misdrijven_2022_2023_clean <- left_join(
  misdrijven_2022_2023,
  vermogen_2023_clean %>% select(gemeenten, huishoudens_x1000) %>% rename(huishoudens_2023 = huishoudens
  by = c("Regio's" = "gemeenten")
) %>%
  left_join(
    vermogen_2022_clean %>% select(gemeenten, huishoudens_x1000) %>% rename(huishoudens_2022 = huishoude
    by = c("Regio's" = "gemeenten")
  ) %>%
  rename(
    misdrijven_2022 = aantal...2,
    misdrijven_2023 = aantal...3
  ) %>%
  mutate(
    misdrijven_per_1000_2023 = as.numeric(misdrijven_2023) / huishoudens_2023,
    misdrijven_per_1000_2022 = as.numeric(misdrijven_2022) / huishoudens_2022
  )
```

Here we extend the same approach to all the three years and join the household counts and crime figures in one data set.

```
misdrijven_2022_2024_clean <- left_join(
  misdrijven_2022_2023,
  vermogen_2023_clean %>% select(gemeenten, huishoudens_x1000) %>% rename(huishoudens_2023 = huishoudens
  by = c("Regio's" = "gemeenten")
) %>%
  left_join(
    vermogen_2022_clean %>% select(gemeenten, huishoudens_x1000) %>% rename(huishoudens_2022 = huishoude
    by = c("Regio's" = "gemeenten")
  ) %>%
  left_join(
    misdrijven_2024_clean %>% select(gemeenten, `Misdrijven 2024`),
    by = c("Regio's" = "gemeenten")
  ) %>%
  left_join(
    vermogen_2024_clean %>% select(gemeenten, huishoudens_x1000) %>% rename(huishoudens_2024 = huishoude
    by = c("Regio's" = "gemeenten")
  ) %>%
  rename(
    misdrijven_2022 = aantal...2,
    misdrijven_2023 = aantal...3
  ) %>%
  mutate(
    misdrijven_per_1000_2022 = as.numeric(misdrijven_2022) / huishoudens_2022,
    misdrijven_per_1000_2023 = as.numeric(misdrijven_2023) / huishoudens_2023,
    misdrijven_per_1000_2024 = as.numeric(`Misdrijven 2024`) / huishoudens_2024
  )
```

3.3 Visualize temporal variation

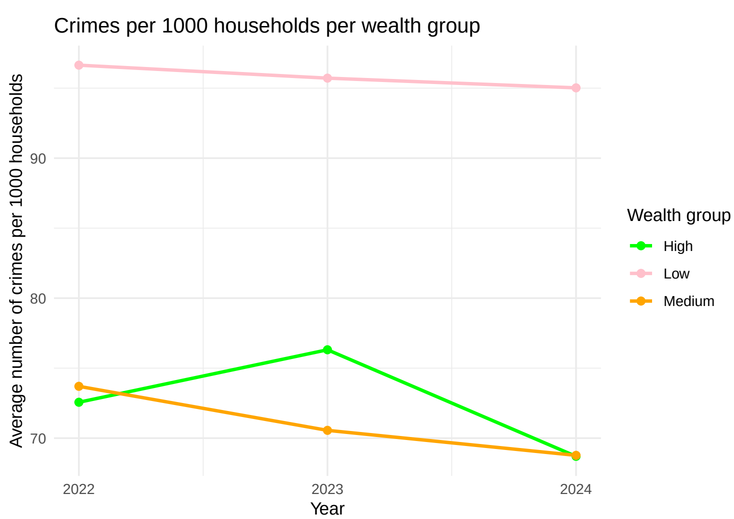
Line plot misdrijven

```
misdrijven_lang <- misdrijven_2022_2024_clean %>%
  select(`Regio's`, starts_with("misdrijven_per_1000")) %>%
  pivot_longer(
    cols = starts_with("misdrijven_per_1000"),
    names_to = "jaar",
    names_prefix = "misdrijven_per_1000_",
    names_transform = as.integer,
    values_to = "misdrijven_per_1000"
  ) %>%
  left_join(
    vermogen_2024_var1 %>% select(gemeenten, vermogen_groep),
    by = c("Regio's" = "gemeenten")
  ) %>%
  filter(!is.na(vermogen_groep))

misdrijven_samenvatting <- misdrijven_lang %>%
  group_by(vermogen_groep, jaar) %>%
  summarise(gem_misdrijven = mean(misdrijven_per_1000, na.rm = TRUE), .groups = "drop")

ggplot(misdrijven_samenvatting,
```

```r
      aes(x = jaar, y = gem_misdrijven, colour = vermogen_groep)) +
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
  scale_x_continuous(breaks = c(2022, 2023, 2024)) +
  scale_colour_manual(
    values = c(
      "Low" = "pink",
      "Medium" = "orange",
      "High" = "green"
    )
  ) +
  labs(
    title = "Crimes per 1000 households per wealth group",
    x = "Year",
    y = "Average number of crimes per 1000 households",
    colour = "Wealth group"
  ) +
  theme_minimal()
```



This plot is relevant to our social problem because it visually shows the direct relationship between income level (low, medium, high) and recorded crime rates in Dutch municipalities from 2022-2024. This graph shows that the low-income group has a significantly higher number of crimes per 1,000 inhabitants, while the middle-income and high-income have significantly lower crime rates.

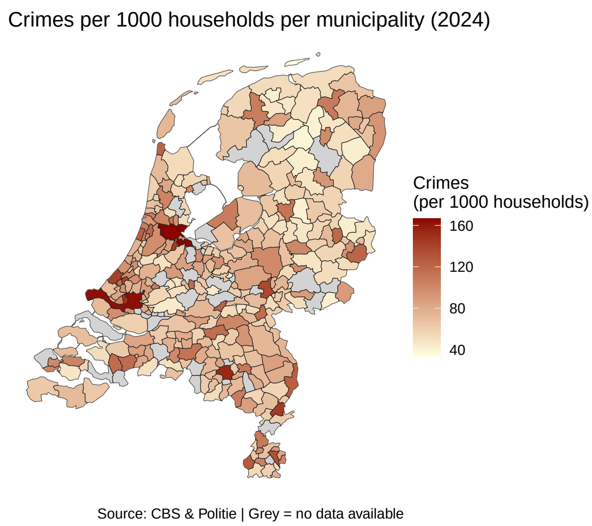
3.4 Visualize spatial variation

Map misdrijven data

```r
#install.packages("cbsodataR")
#install.packages("sf")

library(cbsodataR)
library(sf)
library(tidyverse)
```

```r
misdrijven_gemeenten <- misdrijven_2022_2024_clean %>%
  filter(!grepl("RE|Buitenland|niet in te delen|Nederland|Bron",
                `Regio's`, ignore.case = TRUE))
misdrijven_gemeenten <- misdrijven_gemeenten %>%
  mutate(`Regio's` = case_when(
    `Regio's` == "'s-Gravenhage (gemeente)" ~ "'s-Gravenhage",
    `Regio's` == "Groningen (gemeente)"     ~ "Groningen",
    TRUE                                    ~ `Regio's`
  ))
gemeenten_kaart <- cbs_get_sf("gemeente", 2024)
kaart_data <- gemeenten_kaart %>%
  left_join(misdrijven_gemeenten, by = c("statnaam" = "Regio's"))
ggplot(kaart_data) +
  geom_sf(aes(fill = misdrijven_per_1000_2024)) +
  scale_fill_gradient(low = "lightyellow", high = "darkred", na.value = "lightgrey") +
  labs(
    title = "Crimes per 1000 households per municipality (2024)",
    caption = "Source: CBS & Politie | Grey = no data available",
    fill = "Crimes\n(per 1000 households)"
  ) +
  theme_void()
```



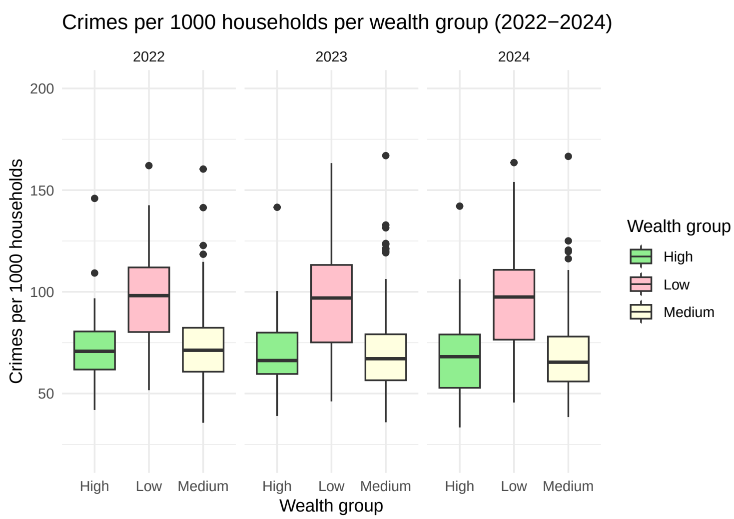
This map is relevant to our social problem because it visualizes the geographic distribution of crime rates across Dutch municipalities. This is an important variable in our hypothesis linking income inequality to crime. By showing where crime rates are the highest and lowest, the map makes a visual comparison with income inequality data at the same municipal level which will contribute to our social problem analysis.

3.5 Visualize sub-population variation

Comparison misdrijven and vermogen per municipality

```r
ggplot(misdrijven_lang,
       aes(x = vermogen_groep, y = misdrijven_per_1000, fill = vermogen_groep)) +
  geom_boxplot() +
```

```r
  facet_wrap(~ jaar) +
  scale_fill_manual(values = c(
    "Low"      = "pink",
    "Medium" = "lightyellow",
    "High"      = "lightgreen"
  )) +
  coord_cartesian(ylim = c(20,200)) +
  labs(
    title = "Crimes per 1000 households per wealth group (2022-2024)",
    x = "Wealth group",
    y = "Crimes per 1000 households",
    fill = "Wealth group"
  ) +
  theme_minimal()
```

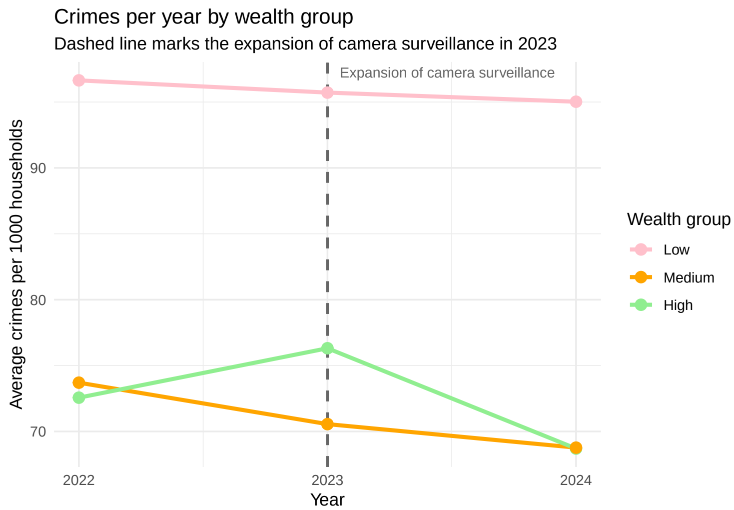


This boxplot is relevant to our social problem because it directly compares crime rates across municipalities grouped by wealth level (low, medium, high). This allows us to see whether lower wealth groups are associated with a higher crime rate compared to the medium and high wealth groups. This supports our hypothesis that different wealth groups are linked to a different amount of crime.

3.6 Event analysis

```r
misdrijven_lang %>%
  filter(!is.na(vermogen_groep)) %>%
  group_by(jaar, vermogen_groep) %>%
  summarise(gemiddeld = mean(misdrijven_per_1000, na.rm = TRUE), .groups = "drop") %>%
  mutate(vermogen_groep = factor(vermogen_groep, levels = c("Low", "Medium", "High"))) %>%
  ggplot(aes(x = jaar, y = gemiddeld, colour = vermogen_groep)) +
  geom_vline(xintercept = 2023, linetype = "dashed", colour = "gray40", linewidth = 0.8) +
  annotate("text", x = 2023.05, y = Inf, label = "Expansion of camera surveillance",
           hjust = 0, vjust = 1.5, size = 3, colour = "gray40") +
  geom_line(linewidth = 1.2) +
  geom_point(size = 3) +
  scale_colour_manual(values = c("Low" = "pink", "Medium" = "orange", "High" = "lightgreen")) +
```

```
  scale_x_continuous(breaks = c(2022,2023,2024)) +
  labs(
    title = "Crimes per year by wealth group",
    subtitle = "Dashed line marks the expansion of camera surveillance in 2023 ",
    x = "Year",
    y = "Average crimes per 1000 households",
    colour = "Wealth group"
  ) +
  theme_minimal()
```



This event analysis examines the impact of increased camera surveillance in the Netherlands in 2023 on crime rates. In recent years, many municipalities have expanded the use of surveillance cameras to improve public safety, deter criminal activity, and support law enforcement in identifying offenders (Koole, 2023). The expectation is that increased surveillance may reduce crime by increasing the perceived risk of being caught. However, it is also possible that certain crimes become more frequently reported or detected due to the presence of cameras. By analyzing crime data around this event, we can assess whether the increase in camera surveillance is associated with a significant change in crime levels in Dutch municipalities (Van de Ven, 2023).

Part 4 - Discussion

4.1 Discuss your findings

The findings of this report suggest that municipalities with lower household wealth generally experience higher levels of registered crime per 1,000 households. This relationship is visible across multiple years (2022,2023,2024) and remains relatively stable over time.

Some limitations should be acknowledged. First, the analysis only uses registered crimes and therefore does not capture unreported crime. Second, household wealth is only one measure of socioeconomic status and does not fully represent inequality. Finally, this analysis covers a period of only three years, which limits the ability to distinguish long-term trends from short-term fluctuations.

Part 5 - Reproducibility

5.1 Github repository link

Public repository: https://github.com/Sophie-Thoolen/Group-1-7---Criminaliteit-en-

5.2 Reference list

Flight, S. (2023, 8 augustus). Gemeentelijk cameratoezicht groeit maar door. Secondant. https://hetccv.nl/secondant-gemeentelijk-cameratoezicht-groeit-maar-door/

Van de Ven, E. (2023, 12 juni). Onderzoek 2023: Momenteel 313.354 beveiligingscamera's in Nederland. VPNGids.nl. https://www.vpngids.nl/onderzoek/onderzoek-2023-momenteel-313354-beveiligingscameras-in-nederland/

Nieuwbeerta, P., McCall, P. L., Elffers, H., & Wittebrood, K. (2008). Neighborhood characteristics and individual homicide risks: Effects of social cohesion, confidence in the police, and socioeconomic disadvantage. Homicide Studies, 12(1), 90–116. https://doi.org/10.1177/1088767907310913